Xiyan Zhou

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EDUCATION

University of Michigan, Ann Arbor Ann Arbor, MI

B.S. Computer Science & B.S. Mathematics (Double Major) *Expected Graduation: December 2026*

GPA: 3.8 / 4.0

Coursework: Measure Theory, Convex Optimization, Bayesian Machine Learning, Numerical Linear Algebra, Real Analysis, Probability Theory, Stochastic Processes, Machine Learning, Operating Systems, Data Structures & Algorithms, Combinatorics, Discrete Mathematics

SKILLS

Programming Languages: Python, C++, C, Java, SQL, MATLAB, JavaScript
Frameworks & Tools: JAX, Flax, PyTorch, NumPy, scikit-learn, LaTeX, Git, Bash, Linux
ML & Math: Reinforcement Learning, Bayesian Methods, Convex Optimization, CNNs, Statistical Modeling
Languages: Mandarin (native), English (proficient)

RESEARCH

Motion Planning via Contrastive Reinforcement Learning

Research Assistant — advised by Kellen Kanarios (Lei Ying Lab) *Jan 2026 – Present*

- Studying goal-conditioned contrastive RL (CRL) with a Gumbel-Softmax actor for discrete action spaces, extending JaxGCRL to categorical logits; diagnosed and fixed a critical action-encoding bug, raising eval success from 3% to 85% (peak 91%) on gridworld tasks.
- Extended CRL to abstract, position-invariant inventory-collection goals via thermometer encoding, reaching 68% success with graceful difficulty degradation and zero-shot generalization to out-of-distribution moving-goal targets.
- Designed a controlled ablation testing a claim from a recent ICML 2026 paper (LEO) that CRL "cannot natively handle" procedurally generated, abstract-goal settings; found that disjunctive goals degrade CRL performance monotonically in the number of satisfiers, a pattern robust across loss configurations and replicated across two environments.
- Implementing double Q-learning, with a hierarchical extension planned; benchmarking agents across Navix and custom grid-world environments, targeting a preprint submission.

AI-Driven Human Digital Twin Study (IOE SURE Program)

Undergraduate Research Assistant — advised by Prof. Jingwen Hu *May 2026 – Present*

- Preprocessed and standardized 45 clinical CT scans into ML-ready left/right leg volumes; built an automated pipeline (cropping, sagittal mirroring, boundary-overlap resolution) for nnU-Net ingestion
- Trained and validated nnU-Net segmentation models achieving 0.92 mean Dice (posterior compartment) and 0.84 mean Dice (medial compartment), with cross-sectional area error within 2–3% against hand-labeled ground truth
- Demonstrated sex-agnostic generalizability: model trained exclusively on female data achieved 0.90 Dice on an independent male test case, comparable to female-only performance (0.92) — informing unified vs. subgroup-specific HERMES parameterization

PROJECTS

Replicating and Extending SimGRL: Robustness in Visual RL under Dynamic Scene Perturbations

- Replicated the SimGRL visual RL framework (PyTorch, DMControl) to study observational overfitting in visual RL agents under dynamic scene perturbations (e.g., shifting distractors and backgrounds).
- Designed four data augmentation extensions (Irregular-SRO, Multi-Source SRO, Hybrid-SRO, Mask-Guided) and ran ablations on the Great Lakes HPC cluster; Hybrid-SRO significantly reduced performance variance versus baseline, yielding a more stable training protocol for high-dimensional visual control.

A Diagnostic Study of Sparsity-Promoting ADMM for Multi-Obstacle Trajectory Optimization

- Formulated an $\ell_{2,1}$ group-sparsity trajectory optimization problem for a 2D double-integrator navigation task with non-convex obstacle-avoidance constraints, and derived a two-block ADMM splitting with over-relaxation and adaptive (residual-balancing) penalty scaling.
- Discovered and diagnosed a structural failure mode: the obstacle projection applied to the auxiliary variable Z does not reliably transfer safety to the dynamically feasible trajectory X — confirmed across a fixed environment, a λ sweep, an iteration-budget sweep (50–1000 iterations), and 10 randomized Monte Carlo trials, where Z was collision-free far more often than X.

Max-Value Entropy Search vs. Expected Improvement for Bayesian Optimization

- Implemented a Gaussian-process-based Bayesian optimization pipeline (GP hyperparameter fitting via marginal-likelihood maximization, BIC-based kernel selection, and an Expected Improvement acquisition function), then re-implemented Max-Value Entropy Search (Wang & Jegelka, ICML 2017) from the literature.
- Found that EI significantly outperformed my MES implementation in a paired comparison over 20 runs ($p = 1.32 \times 10^{-3}$); diagnosed the cause as a small fitted GP lengthscale that made the objective resemble noise, undermining MES's reliance on estimating the posterior distribution over the optimum.

AWARDS & TEACHING

- **James B. Angell Scholar** — University of Michigan (March 2025); awarded to students with a perfect 4.0 term GPA
- **University Honors** — University of Michigan (×4: Fall 2023, Winter 2024, Fall 2024, Winter 2025)
- **Grader** — MATH 571 (Graduate Numerical Linear Algebra) and MATH 371, EECS 203
- **Peer Tutor**, Math Learning Center (4 semesters)